MRCNet: Motion Reasoning Chain for Cross Modal Video Camouflaged Object Detection

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***Abstract*—Video camouflaged object detection (VCOD) aims to identify objects that seamlessly blend into their surroundings in video sequences. Traditional methods merely rely on visual cues to capture inter-frame motion that reveals camouflaged objects. However, the high similarity between camouflaged objects and their environments often renders pure reliance on visual cues unreliable. Additionally, random motions including camera shaking and abrupt scene transitions also inevitably bring noise into the identification process. To overcome these challenges, we propose a Motion Reasoning Chain Network (MRCNet), a novel cross-modal VCOD framework that emulates the human thought process when observing camouflaged objects, i.e., motion reasoning. Specifically, we introduce a generative sampling strategy grounded in multimodal large language models (MLLMs) to bridge the implicit knowledge space of MLLMs and the explicit representation space regarding the attributes of camouflaged objects, thereby enabling the effective establishment of the motion reasoning chain tailored for VCOD. This process provides semantic guidance for visual comprehension of camouflaged objects through motion and concept attribute reasoning. To improve the identification capability of camouflaged objects, we develop motion representation learning driven by the motion reasoning chain. It introduces hierarchical de-biased motion prototype learning to mitigate hallucinations of MLLMs, boosting the motion perception. To learn precise prompts for the visual foundation model, cross-modal prompt learning further incorporates the de-biased concept prototype into visual representations to enhance the visual comprehension of camouflaged objects. Extensive experiments across three datasets demonstrate that MRCNet achieves state-of-the-art results on both general metrics and spatiotemporal consistency metrics.**

***Index Terms*—Camouflaged object detection, motion reasoning chain, prompt learning, multimodal large language model.**

Received 8 April 2025; revised 27 March 2026; accepted 24 April 2026. Date of publication 1 May 2026; date of current version 6 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62476022 and Grant 62506029, in part the Beijing Nova Program under Grant 202504841007, and in part by the Beijing Natural Science Foundation under Grant L251043 and Grant 4254085. Recommended for acceptance by Z. Li. *(Corresponding author: Zhenfeng Zhu.)*

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Digital Object Identifier 10.1109/TPAMI.2026.3689767

I. Introduction

Camouflaged object detection (COD) [1] is a crucial task in computer vision, concentrating on identifying objects that seamlessly blend into their surroundings. The subtle indistinguishability between camouflaged objects and their environments makes COD more challenging than traditional object detection [2], [3], [4]. The advance of the COD technique has been proven to be beneficial to a wide range of applications, including military [5] and security [6], wildlife conservation [7], and medical image analysis [8], [9]. While significant progress has been made in COD for static images [10], [11], recent studies [12], [13] indicate that human beings are born with a distinctive perception ability of camouflaged objects in motion. In fact, this phenomenon is consistent with the two-stream hypothesis in neuroscience [14], which posits that the human visual cortex operates with a ventral stream for recognition and a dorsal stream for motion processing. This insight has sparked a growing interest in video camouflaged object detection (VCOD) [15].

Previous VCOD methods, as illustrated in Fig. 1(a), model the motion between frames implicitly or explicitly to detect camouflaged objects. With the emergence and development of visual foundation large models [16], temporal-spatial prompt learning [17] in Fig. 1(b) is introduced to excavate their capabilities in VCOD. Despite exhibiting promising results, they still struggle with the precise segmentation of camouflaged objects in complex scenarios characterized by low light, background interference, and occlusion. This can be mainly attributed to two aspects. First, these methods purely rely on visual cues, while the high similarity between camouflaged objects and their environments makes visual information often unreliable. Second, the motion in video sequences is compounded by the ego-motion of camouflaged objects and random motions, including camera shaking, scene changes, etc. These underlying random motions inevitably interfere with the identification of camouflaged objects. Taking these into consideration, we summarize the challenge confronted with the VCOD task from two perspectives: 1) *How to suppress the interference from underlying random motions and infer the location of camouflaged objects?* 2) *How to suppress the obvious interference from the background and mine subtle discriminative cues of camouflaged objects more reliably?*

In neuroscience, when observing camouflaged objects, the human reasoning is considered to be a multi-step process [18]. First, the brain uses previous knowledge to quickly understand the general context. Second, it further infers the potential object

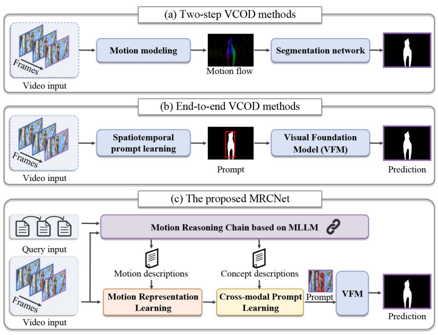


Fig. 1. Illustrations of existing VCOD methods. (a) Two-step VCOD; (b) End-to-end VCOD; (c) Our MRCNet model.

categories. Then, motion perception is used to distinguish the motion patterns of the background and the object, and random motions in the background are excluded. As the object location is confirmed, subtle visual cues are integrated to perform pixel-level object detection. By mimicking this process, we can design a thought-reasoning strategy to improve the accuracy and robustness of COD.

With this inspiration, we present a novel **M**otion **R**easoning **C**hain **Net**work, abbreviated by **MRCNet** hereafter, for cross-modal VCOD. As presented in Fig. 1(c), we imitate the human thought reasoning to establish a motion reasoning chain to guide visual comprehension of camouflaged objects. Using the prior knowledge of multimodal large language models (MLLMs), on the one hand, the camouflaged object is accurately located by reasoning about the motion semantics, and on the other hand, the discriminative cues and high-confidence pixels are mined by the potential concept semantics. **To the best of our knowledge**, it is the first attempt to explore the MLLM-based motion reasoning chain to guide visual comprehension of camouflaged objects, paving a new way for camouflaged object detection. Specifically, our main contributions can be highlighted in the following aspects:

- Innovatively, we introduce an MLLM based generative sampling strategy to establish the bridge between the implicit knowledge space of the MLLM and explicit representation space concerning the attributes of camouflaged objects, thereby facilitating the construction of the motion reasoning chain for COD.
- To promote the motion perception of camouflaged objects, the MRC-driven motion representation learning is proposed, in which a hierarchical de-biased motion prototype learning is introduced to address the semantic drifts caused by the hallucination of the MLLM.
- To learn precise prompts for the visual foundation model, cross modal prompt learning that integrates de-biased concept semantic prototype into visual representations is adopted to promote the visual comprehension of camouflaged objects.
- Without bells and whistles, extensive experiments on three benchmarks demonstrate that MRCNet achieves substantial improvements in both general metrics and our proposed spatiotemporal consistency metrics.

The remainder of this paper is organized as follows. The review of related work on camouflaged object detection is introduced in Section II. In Section III, we present in detail the methodology of the proposed MCRNet. The experiments are given in Section IV. Finally, we give our conclusions and future work in Section V.

II. Related Works

A. Camouflaged Object Detection

Camouflaged object detection (COD) [19], [20], [21], [22] aims to identify objects that blend into their surroundings. Existing methods mainly follow three strategies: (i) Multi-stage learning: This strategy refines object detection results through sequential stages. For instance, SINet [1] and SINet-V2 [23] introduced a two-stage framework, consisting of an initial localization stage followed by a segmentation stage. PENet [24] further added a boundary restoration stage to enhance the edge precision. (ii) Multi-task learning: This category [25], [26] incorporates auxiliary tasks, such as classification, localization, and edge detection, to derive shared context representations. (iii) Bionic strategy: This line of work [27], [28] mimics human zooming by processing input images at multiple scales.

With the recent advancements in MLLMs [29], [30], cross-modal interaction has been increasingly explored for COD. For instance, GenSAM [31] used a chain of thought for static semantic reasoning on visual prompts. In contrast, our MRCNet reasons about motion and concept attributes via the motion reasoning chain to guide the visual comprehension of camouflaged objects, resulting in more accurate prompts.

B. Video Camouflaged Object Detection

Unlike static images, video sequences offer a key advantage, i.e., the presence of motion. Existing video camouflaged object detection (VCOD) methods [12], [15], [17], [28], [32], [33], [34] typically model motion across frames explicitly or implicitly for breaking camouflage. Explicit motion-based methods [12], [32] segmented the optical flow field after motion compensation or frame registration to distinguish objects from the background. However, the inaccuracy in optical flow estimation leads to frame misalignment, degrading detection accuracy. In contrast, implicit motion-based methods utilized neural networks to learn feature correspondences across frames for better alignment [15], [17], [28], [34]. While alignment issues are mitigated, their pure reliance on visual modality restricts their ability to handle more complex and advanced camouflage. Different from the above methods, the proposed MRCNet fully explores the complementary advantages of visual perception and textual semantics, enhancing the comprehension of the visual foundation model on camouflaged objects.

C. Segment Anything Model

The segment anything model (SAM) [16], a prompt-driven visual foundation model, has demonstrated unprecedented performance in natural image segmentation tasks. Based on the impressive representations learned on the huge SA-1B dataset [16],

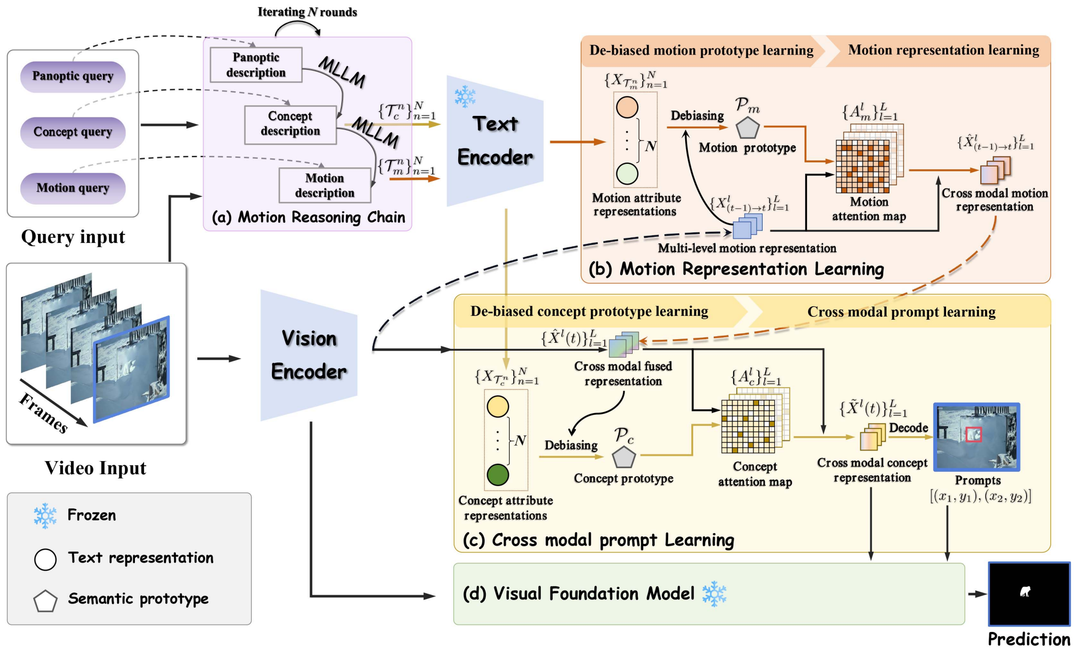


Fig. 2. Framework of the proposed MRCNet. This sequential pipeline leverages an MLLM-based motion reasoning chain to infer motion and concept attributes, providing semantic guidance to progressively enhance the motion perception and visual comprehension of camouflaged objects. It comprises four key components, i.e., (a)–(d). During the training phase, the visual foundation model is frozen, including the image encoder, prompt encoder, and mask decoder.

SAM has exhibited excellent zero-shot segmentation capability across diverse scenarios [35], [36]. Despite its successes [37], SAM often encounters unique challenges when applied to camouflaged object detection, where camouflaged objects are visually indistinguishable from the background, making it particularly difficult to obtain accurate human-provided prompts. To reduce SAM's reliance on human-provided prompts, TSP-SAM [17] leveraged spatiotemporal relationships across frames to learn visual prompts for SAM. In contrast to the pure reliance on visual cues in TSP-SAM, the proposed MRCNet emphasizes cross modal learning by a motion reasoning chain. This chain infers both motion and concept attributes of camouflaged objects in video sequences, offering semantic guidance for visual comprehension of camouflaged objects.

D. Multimodal Large Language Models

The strong generalization ability of large language models (LLMs) [38] has driven the rapid progress of visual foundation models (VFMs) [16], [39] and multimodal large language models (MLLMs) [29], [30]. By jointly modeling visual and textual modalities, MLLMs enable unified decoding of image and text tokens and exhibit strong adaptability across vision–language tasks. Representative models, including LLaVA [30], Qwen-VL [40], and InternVL [41], demonstrate the effectiveness of large-scale multimodal pretraining and achieve impressive zero-shot and few-shot generalization.

To improve the performance of MLLMs on fine-grained understanding tasks [42], [43], existing studies [44], [45], [46] adopt instruction tuning to align language descriptions with spatial regions, typically reformulating referential comprehension datasets into an instruction-following format or leveraging grounding models [47] to generate instruction tuning data. However, such data often covers only a limited set of referential comprehension tasks, restricting generalization beyond the tuned scenarios. In contrast, our work aims to activate the inherent perceptual capability of MLLMs in camouflaged scenes via a motion reasoning chain, providing semantic guidance for fine-grained visual comprehension of camouflaged objects.

III. Methodology

A. Framework Overview

The overall framework of the proposed MRCNet is illustrated in Fig. 2, which mainly consists of four key modules:

- The **Motion reasoning chain** (MRC) uses a multimodal large language model to recursively infer the motion and concept attributes of camouflaged objects, providing semantic guidance for visual comprehension.
- The **MRC-driven motion representation learning** (MRL) takes a hierarchical de-biased motion prototype learning strategy to mitigate the hallucination effects in the motion attributes, and then combines the learned motion prototype with the visual motion representations to strengthen the motion perception of camouflaged objects.
- The **Cross-modal prompt learning** (CPL) combines the de-biased concept semantic prototype with visual representations to enhance visual comprehension of camouflaged objects and improve the precision of prompts.
- The **visual foundation model** with semantic knowledge injection (SKI) receives video frames alongside prompts

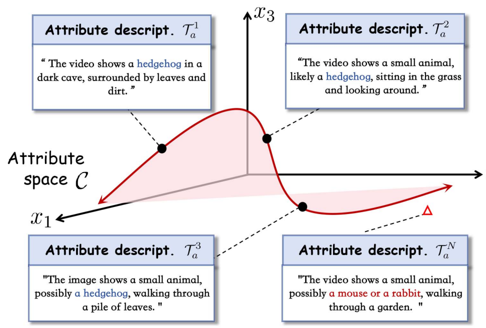


Fig. 3. Illustration of the generative sampling. ● represents a discrete sample in the continuous attribute space. △ represents outliers due to the hallucinations of MLLMs.

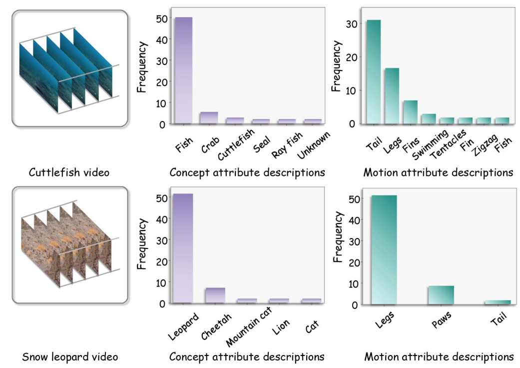


Fig. 4. Frequency distribution of discrete concept and motion sampling points of camouflaged objects.

and cross modal concept representations as inputs to predict pixel-level segmentation masks.

B. Motion Reasoning Chain Based on MLLM

Traditional VCOD methods [15], [17], [28], [33] primarily rely on visual cues, which often fail due to the inherent similarity between camouflaged objects and their surroundings. Additionally, uncertain motions in video sequences, such as scene changes and camera shaking, further interfere with the identification of camouflaged objects. To address these limitations, it is necessary to introduce prior semantic knowledge to guide visual comprehension of camouflaged objects.

MLLM based generative sampling: Due to the extensive knowledge spanning diverse scenarios, multimodal large language models (MLLMs) exhibit advanced cross modal understanding and reasoning capabilities. Assuming that the attributes of camouflaged objects are embedded within the prior knowledge of the MLLM in an implicit manner, it is non trivial to directly model this latent space due to its complexity and unobservability. As an alternative way, we propose an MLLM based generative sampling strategy to transfer the implicit continuous knowledge space into explicit discrete representation space, thereby modeling the attribute distribution. To this end, we have the following proposition:

Proposition 1. (Generative Sampling): Let $\mathcal{C}$ represent the implicit continuous knowledge space of the MLLM regarding the set $\mathcal{A}$ of attributes of camouflaged objects, and $\mathcal{F}_{\text{MLLM}}$ be the MLLM-based generative sampler. The discrete sampling process can be defined as: $\{\mathcal{C}_a\} \xrightarrow{\mathcal{F}_{\text{MLLM}}} \{\mathcal{T}_a^1, \ldots, \mathcal{T}_a^N\}$, where N denotes the number of sampling, and $a \in \mathcal{A}$ is the specified attribute.

Essentially, this process builds a bridge between the implicit continuous attribute knowledge space of MLLMs and the explicit discrete attribute representation space. In a sense, the generation process of MLLM could be interpreted as *generative sampling*. As shown in Fig. 3, the generated set $\mathcal{T}_a = \{\mathcal{T}_a^1, \ldots, \mathcal{T}_a^N\}$ comprises discrete sampling points supporting the attribute subspace $\mathcal{C}_a$. The diversity of these samples stems from the inherent generative variety of MLLMs. Notably, due to the hallucinations of MLLMs, some certain samples may also be generated as outliers, such as the samples marked by △.

Without loss of generality, Fig. 4 presents two representative cases of generative sampling results, in which MLLMs generate the motion and concept attribute descriptions of camouflaged objects with the video input and predefined query (Prompt). By analyzing the frequency distribution of the discrete sampling point set $\mathcal{T}_a$ of concept and motion attributes, it is evident that the hallucination inevitably appears. For instance, in the case of a video depicting a cuttlefish, concept attribute sampling yields correct samples such as 'Fish' and 'Cuttlefish', but also produces semantically incorrect outliers such as 'Crab' and 'Seal'. Similarly, the motion attribute like 'Tail' and 'Fins' are mixed with irrelevant terms like 'Swimming' and 'Zigzag', which fail to describe the motion cues for object localization precisely. These outliers introduce semantic ambiguity and bias, potentially degrading the reliability of subsequent visual understanding of camouflaged objects.

Motion reasoning chain: As shown in Fig. 5, given a video sequence $V \in \mathbb{R}^{T\times H\times W\times 3}$, the motion reasoning chain recursively infers three key attributes of camouflaged objects, including panoptic (p), concept (c), and motion (m). Formally, the attribute set is expressed as $\mathcal{A} = \{p, c, m\}$ according to Proposition 1. Note that the predefined queries corresponding to these attributes are adopted to guide the explicit discretization of the implicit continuous knowledge space of the MLLM, activating the generative sampling process.

Specifically, for the n-th round of the motion reasoning chain via generative sampling mentioned above, it is constituted of the following sequential steps:

Step 1: Inspired by the generated knowledge prompting [48], an initial panoptic attribute is sampled to depict the overall camouflage scenario:

$$\mathcal{T}_p^n = \mathcal{F}_{\text{MLLM}}(V, Q_p) \tag{1}$$

where the panoptic query Q_p is defined as *'Describe the movement of the camouflaged animal in this video shortly'* and $\mathcal{F}_{\text{MLLM}}$ is implemented using Video-LLaVA [49].

Step 2: Building upon this panoptic description $\mathcal{T}_p^n$, the potential concept attribute of camouflaged objects in this scenario is further sampled via:

$$\mathcal{T}_c^n = \mathcal{F}_{\text{MLLM}}\left(V, \mathcal{T}_p^n, Q_c\right) \tag{2}$$

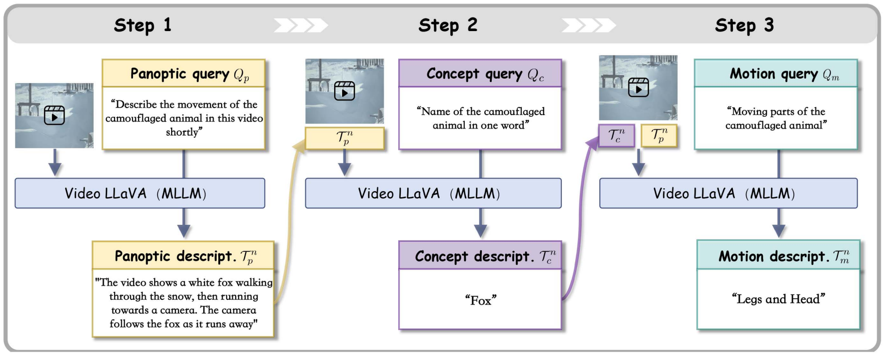


Fig. 5. Pipeline of the proposed motion reasoning chain. The $\mathcal{T}_p^n$, $\mathcal{T}_c^n$, and $\mathcal{T}_m^n$ are the panoptic, concept, and motion descriptions obtained at the n-th round of the motion reasoning chain, respectively.

where the concept query Q_c is formulated as *'Name of the camouflaged animal in one word'*. Note that the concept description $\mathcal{T}_c^n$ characterizes well the appearance details of camouflaged objects.

Step 3: Taking $\mathcal{T}_p^n$ and $\mathcal{T}_c^n$ as recursive inputs, the motion attribute is sampled to delineate the local changes of camouflaged objects. This process is represented as follows:

$$\mathcal{T}_m^n = \mathcal{F}_{\text{MLLM}}\left(V, \mathcal{T}_p^n, \mathcal{T}_c^n, Q_m\right) \tag{3}$$

where the motion query is designed as *'Moving parts of the camouflaged animal'*.

Through the sequential sampling in the motion reasoning chain, discrete motion and concept attribute descriptions represented as $\{\mathcal{T}_c^n\}_{n=1}^N$ and $\{\mathcal{T}_m^n\}_{n=1}^N$ are derived to guide the motion analysis and visual understanding of camouflaged objects.

C. MRC-Driven Motion Representation Learning

The accurate identification of camouflaged objects is often hindered by underlying random motions in video sequences. To suppress these random motions and enhance the motion perception of camouflaged objects, it is necessary to leverage prior motion semantics derived from the motion reasoning chain to refine the motion representations for more accurate object recognition.

1) Visual Motion Perception: Given consecutive video frames $V(t-1), V(t) \in \mathbb{R}^{H\times W\times 3}$, $t \in \{1,\ldots,T\}$, following ZoomNeXt [28], we adopt a vision encoder that comprises a shared triplet feature extraction network, a scale merging sub-network, and a rich granularity perception unit to extract visual representations $X^l(t-1), X^l(t) \in \mathbb{R}^{H_l\times W_l\times C_l}$, where $l \in \{1,\ldots,L\}$ denotes the level of the vision encoder blocks, $H_l = H/2^{l+1}$ and $W_l = W/2^{l+1}$ specify the resolution, and C_l is the number of channels.

To perceive the changes between consecutive frames, the multi-level motion representations are derived as follows:

$$X^l_{(t-1)\to t} = \text{LN}\left(X^l(t) + \text{MHCA}(X^l(t), X^l(t-1))\right) \tag{4}$$

where MHCA$(\cdot,\cdot)$ denotes the multi-head cross attention [50], and LN$(\cdot)$ is layer normalization. Note that the motion representation $X^l_{(t-1)\to t}$ captures the inter-frame correspondences within the visual space and characterizes the compound motions in video sequences.

2) De-Biased Motion Prototype Learning: It is well-known that MLLMs are susceptible to hallucination effects, yielding uncertain reasoning outcomes [51]. We have illustrated the frequency distribution of the discrete attribute sampling points from the motion reasoning chain as in Fig. 4. In common sense, two key issues can be observed: (i) **Semantic redundancy.** The discrete sample set often contains multiple near-duplicate or semantically overlapping samples, resulting in an inefficient semantic representation space. (ii) **Hallucinated outliers.** Due to the inherent uncertainty in MLLMs, some samples deviate from the true attribute semantics, introducing semantic ambiguity and bias into the subsequent cross-modal understanding. These issues pose an urgent need for robust strategies to filter or rectify misleading generative samples. To deal with them, we propose a hierarchical framework consisting of bottom-up self-evolving semantic condensation with top-down visual-guided semantic aggregation. This framework aggregates discrete motion attribute representations into a robust *motion prototype*, ensuring reliable motion semantic estimation from noisy generative samples.

For discrete motion attribute descriptions $\{\mathcal{T}_m^n\}_{n=1}^N$, to minimize the semantic redundancy and balance efficiency with robustness, a bottom-up self-evolving semantic condensation mechanism as shown in Fig. 6 is introduced to extract representative motion descriptors. Specifically, each sample $\mathcal{T}_m^n$ is embedded into a predefined template *'[Motion attribute] of a camouflaged animal'* and processed by a text encoder, implemented as RoBERTa [52] in our work, yielding motion attribute representations $X_{\mathcal{T}_m^n} \in \mathbb{R}^{N_{token}\times d}$, where N_{token} denotes the number of tokens and d is the representation dimension. Subsequently, a K-means based clustering operation is applied to derive representative motion descriptors $\{\mathcal{D}_m^1,\ldots,\mathcal{D}_m^K\}$.

However, it is also worth noting that due to the presence of outliers, certain motion descriptors may deviate from the true semantic center, affecting the accurate characterization of the motion attribute. To counter this, we employ a top-down

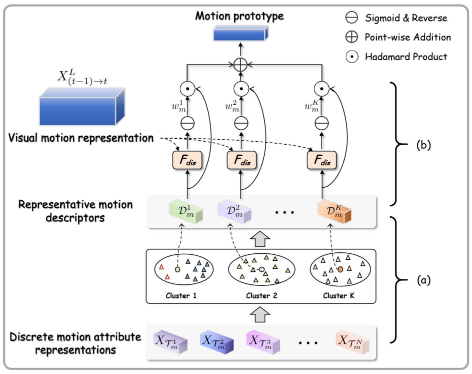


Fig. 6. Illustration of the de-biased motion prototype learning. (a) Bottom-up self-evolving semantic condensation. (b) Top-down visual-guided semantic aggregation.

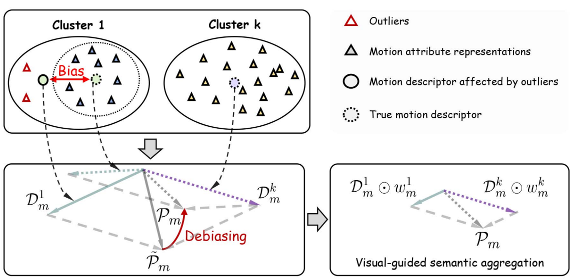


Fig. 7. Illustration of the aggregation of motion descriptors. This process employs adaptive weight assignment with the guidance of visual representations to reduce the importance of the biased descriptor D_m^1, thereby suppressing hallucinations and learning the true motion prototype P_m.

visual-guided semantic aggregation to learn the motion prototype. Formally, it is expressed as follows:

$$\mathcal{P}_m = \sum_{k=1}^{K} (\mathcal{D}_m^k \odot w_m^k) \tag{5}$$

where $\odot$ denotes the broadcast Hadamard product. Note that the importance weight w_m^k is obtained by measuring the distance between the motion representations $X_{(t-1)\to t}^L$ and the k-th motion descriptor $\mathcal{D}_m^k \in \mathbb{R}^{1\times d}$:

$$w_m^k = \mathbf{1} - \text{Sigmoid}\left(F_{dis}\left(X_g \otimes \text{CBN}(X_{(t-1)\to t}^L), \mathcal{D}_m^k\right)\right) \tag{6}$$

where $F_{dis}(\cdot,\cdot)$ denotes the distance metric, $X_g \in \mathbb{R}^{1\times H_L W_L}$ is a learnable matrix to extract the global representation of motion, $\otimes$ denotes matrix multiplication, and $\text{CBN}(\cdot)$ applies convolution, batch normalization, and spatial flattening. Here, the greater the semantic biases of the motion descriptor, the farther the distance between the motion descriptor and the motion representation, resulting in a smaller contribution w_m^k of the motion descriptor to the motion prototype. Taking the motion descriptors $\mathcal{D}_m^1$ and $\mathcal{D}_m^k$ in Fig. 7 as examples, when biased descriptors are aggregated with uniform weights, the resulting prototype $\tilde{\mathcal{P}}_m$ deviates from the true prototype $\mathcal{P}_m$. In contrast, (5) assigns the importance weights w_m^1 and w_m^k with the guidance of visual representations to weaken the interference of the biased descriptor, thereby learning the true motion prototype.

Through this hierarchical debiasing framework, the hallucination artifacts associated with MLLMs are effectively eliminated, and the movement of camouflaged objects is precisely outlined by the motion prototype $\mathcal{P}_m$.

3) Prototype-Induced Motion Representation Learning: To enhance the visual motion perception of camouflaged objects, we leverage the de-biased motion prototype to refine the motion representations, thereby suppressing uncertain motions in video sequences and improving object identification. Specifically, as shown in Fig. 2, the pixel-level correlations between the motion prototype $\mathcal{P}_m \in \mathbb{R}^{1\times d}$ and the flattened motion representations $X_{(t-1)\to t}^l \in \mathbb{R}^{H_l W_l \times C_l}$ are measured via:

$$A_m^l = \text{Sigmoid}\left(\text{MLP}(X_{(t-1)\to t}^l) \otimes \text{MLP}(\mathcal{P}_m)\right) \tag{7}$$

where $\text{MLP}(\cdot)$ denotes a multi-layer perceptron. Note that $A_m^l \in \mathbb{R}^{H_l W_l \times 1}$ represents the motion likelihood of each pixel, highlighting regions associated with camouflaged objects' movement. It acts as an attention map to refine the motion representations:

$$\hat{X}_{(t-1)\to t}^l = \text{LN}\left(X_{(t-1)\to t}^l + A_m^l \odot X_{(t-1)\to t}^l\right) \tag{8}$$

Note that the derived cross modal motion representation $\hat{X}_{(t-1)\to t}^l \in \mathbb{R}^{H_l \times W_l \times C_l}$ significantly promotes both motion perception and accurate identification of camouflaged objects.

D. Cross-Modal Prompt Learning

MRC-driven motion representation learning embeds precise motion semantics within visual motion representations to facilitate the identification of camouflaged objects. To further improve the completeness of the detected objects, incorporating prior concept semantics from MRC is essential for a deeper exploration of subtle visual clues. This enables significantly advancing the understanding of camouflaged objects while learning precise prompts for the visual foundation model.

1) De-Biased Concept Prototype Learning: Analogous to the way of motion prototype learning in Fig. 6, this module aims to address the hallucination effects in concept attributes and achieve robust estimation of concept semantics. For discrete concept attribute descriptions $\{\mathcal{T}_c^n\}_{n=1}^N$, each of them is formatted as *'a photo of a [concept attribute]'*, and processed through the self-evolving semantic condensation process. The extracted representative concept descriptors $\{\mathcal{D}_c^1, \ldots, \mathcal{D}_c^K\}$ capture spatial semantic contexts of potential camouflaged objects while minimizing redundancy and ensuring efficiency.

To further eliminate the semantic biases in concept descriptors and precisely delineate the concept category of camouflaged objects, visual-guided semantic aggregation is employed to align concept semantics with visual representations. Specifically, a cross attention [50] is established to map the cross modal motion representation $\hat{X}_{(t-1)\to t}^l$ into the visual space at frame t via:

$$\hat{X}^l(t) = \text{LN}\left(X^l(t) + \text{MHCA}\left(\hat{X}_{(t-1)\to t}^l, X^l(t)\right)\right) \tag{9}$$

where $l \in \{1, \ldots, L\}$ denotes the level of vision encoder blocks. Guided by the cross modal fused representation $\hat{X}^L(t)$, the concept prototype is derived as follows:

$$\mathcal{P}_c = \sum_{k=1}^{K} \left(\mathcal{D}_c^k \odot w_c^k\right) \tag{10}$$

where $w_c^k = \mathbf{1} - \mathrm{Sigmoid}(F_{dis}(X_g \otimes \mathrm{CBN}(\hat{X}^L(t)), \mathcal{D}_c^k))$ represents the importance weights obtained by reversing the distance between the cross modal fused representation $\hat{X}^L(t)$ and the k-th concept descriptor $\mathcal{D}_c^k$.

2) Concept-Aware Cross-Modal Prompt Learning: To improve the accuracy of prompts, the concept prototype $\mathcal{P}_c$ is integrated into cross modal fused representations $\hat{X}^l(t)$ to enhance the visual comprehension of camouflaged objects. The pixel-level correlations between them are calculated by:

$$A_c^l = \mathrm{Sigmoid}\left(\mathrm{MLP}(\hat{X}^l(t)) \otimes \mathrm{MLP}(\mathcal{P}_c)\right) \tag{11}$$

Here, A_c^l represents the probability that each pixel belongs to this specific concept category and serves as an importance map to enhance the category-specific features within the visual space:

$$\tilde{X}^l(t) = \mathrm{LN}\left(\hat{X}^l(t) + A_c^l \odot \hat{X}^l(t)\right) \tag{12}$$

where $\tilde{X}^l(t)$ comprehensively captures both the motion and concept semantics of camouflaged objects, effectively mining subtle visual clues of camouflaged objects.

To yield precise prompts for the visual foundation model, a decoding process is employed to predict a mask:

$$P_d(t) = E_d\left(\{\tilde{X}^l(t)\}_{l=1}^{L}\right) \tag{13}$$

where E_d refers to the decoder [15]. Note that the final box prompt $P_b(t) = [x_1, y_1, x_2, y_2]$ is derived from the largest connected component of the predicted mask $P_d(t)$, where (x_1, y_1) and (x_2, y_2) denote the coordinates of the top-left and bottom-right corners, respectively. By combining the prior semantic knowledge from the motion reasoning chain with the visual representations, the prompts not only depict the spatial position of camouflaged objects but also delineate the motion relationships within the video sequences, promoting the robust cross modal VCOD.

E. Visual Foundation Model With Semantic Knowledge Injection

It is widely acknowledged that the remarkable performance of the visual foundation model relies not only on well-learned prompts but also on the representational capability of the image encoder. Nevertheless, since the image embedding of the visual foundation model is deficient in semantic information, the final predictions may be ambiguous. Hence, it is necessary to inject the semantic knowledge associated with camouflaged objects into the image embeddings.

As depicted in Fig. 2, the visual foundation model receives the video frame $V(t)$, the box prompt $P_b(t)$, and the cross-modal concept representation $\tilde{X}^L(t)$ from (12) as inputs to predict the final segmentation mask $P(t)$. In the visual foundation model, the process of injecting the semantic knowledge into the image embedding $X_e(t)$ is expressed as:

$$\hat{X}_e(t) = X_e(t) + \mathrm{CBN}(\mathrm{Concat}(X_e(t), \tilde{X}^L(t))) \tag{14}$$

where $\mathrm{Concat}(\cdot)$ means channel-wise concatenation. In this way, through semantic knowledge injection, the visual foundation model enables to fully comprehend the segmented object and achieves more precise camouflaged object detection in videos.

F. Model Optimization

We train the proposed MRCNet model by minimizing the joint loss below:

$$\mathcal{L}(t) = \mathcal{L}_{SAM}(P(t), G(t)) + \alpha\mathcal{L}_{prompt}(P_d(t), G(t)) \tag{15}$$

where α is a balance factor between the prediction loss and prompt loss. $\mathcal{L}_{SAM}$ supervises the final prediction and $\mathcal{L}_{prompt}$ is the supervision on the prompts. They are expressed as:

$$\mathcal{L} = \mathcal{L}_{ce}^{w} + \mathcal{L}_{iou}^{w} + \mathcal{L}_{dice} \tag{16}$$

where $\mathcal{L}_{ce}^{w}$ is the weighted cross-entropy loss, $\mathcal{L}_{iou}^{w}$ is the weighted intersection-over-union loss and L_{dice} is the dice loss. In the prompt loss L_{prompt}, UAL loss [28] is also used to optimize the prompts.

IV. Experimental Results and Analysis

A. Experiment Settings

Datasets: Following previous works [15], [28], we evaluate the proposed MRCNet on three widely-used datasets: MoCA-Mask [15], CAD2016 [53] and CPD [54]. The MoCA-Mask dataset, a video camouflaged object detection dataset, features camouflaged animals moving in natural scenes. It comprises 87 video sequences (totaling 22939 frames) and is divided into 71 sequences for training and 16 sequences for testing. Each video is meticulously annotated with bounding boxes and pixel-level segmentation masks on every fifth frame. The Camouflaged Animal Dataset [53] (CAD2016), sourced from YouTube videos, is a small video camouflaged object detection dataset. It includes nine short video clips, also with pixel-level segmentation masks on every fifth frame. Moreover, CPD [54] is a camouflaged people detection dataset. It comprises 2600 video frames with pixel-level segmentation masks.

Implementation details: Following previous approaches [15], [28], we initially pretrain MRCNet on the COD10 K dataset [23] and then finetune it on the MoCA-Mask dataset. To ensure a fair comparison, both the training and testing images are resized to 352×352. Throughout the training process, the components of the visual foundation model SAM remain frozen. We leverage Adam [55] as our optimizer and set the weight decay to 1×10^{-5}. In the pretraining stage, the learning rate is initialized to 1×10^{-4} for the triplet feature extraction network and 5×10^{-4} for other trainable parameters with a maximum of 30 epochs. During the fine-tuning process on the MoCA-Mask dataset, the learning rate is initialized to 5×10^{-5} for the motion-related parameters and 1×10^{-5} for other parameters. After 8 epochs,

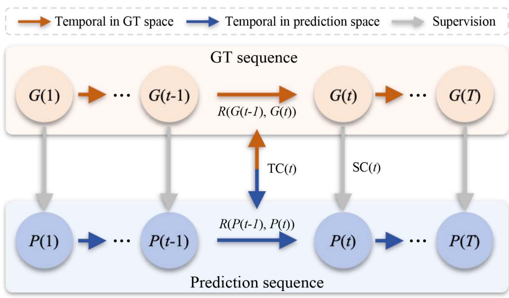


Fig. 8. Illustration of the proposed spatiotemporal consistency metric. $R(\cdot,\cdot)$ represents the temporal Region of Interest (ROI) extractor. $\mathrm{TC}(t)$ and $\mathrm{SC}(t)$ denote the temporal continuity metric and spatial consistency metric, respectively.

the learning rate is reduced by 90%. The CAD2016 and CPD benchmarks are utilized to evaluate the generalization.

B. Evaluation Metrics

1) General Metrics: During the inference phase, we evaluate the performance of the methods using six commonly-used metrics [21]: S-measure (S_α), Weighted F-measure (F_β^w), Enhanced-alignment measure (E_ϕ), Mean absolute error (MAE, $\mathcal{M}$), meanDice (mDice), meanIoU (mIoU). A superior VCOD method would produce higher scores for S_α, F_β^w, E_ϕ, mDice and mIoU, concurrently yielding a lower MAE.

2) Spatiotemporal Consistency Metric: The evaluation metrics discussed above primarily address spatial similarity between the prediction and the ground truth on a frame-by-frame basis. However, these metrics fail to account for inter-frame consistency, which is crucial for video-based tasks. To the best of our knowledge, no existing metric quantitatively evaluates the temporal consistency. To bridge this gap, we propose a new spatiotemporal consistency metric, integrating spatial accuracy and temporal continuity into a unified framework for a comprehensive evaluation, as shown in Fig. 8.

In order to quantify spatial consistency, we employ the general metrics to measure the accuracy of the prediction for each frame. Formally, the spatial consistency for the s-th video sequence at frame t is defined as:

$$\mathrm{SC}_s(t) = Prob(P_s(t), G_s(t)) \triangleq \frac{P_s(t) \cap G_s(t)}{P_s(t) \cup G_s(t)} \tag{17}$$

where $P_s(t)$ and $G_s(t)$ are the prediction and ground truth, respectively. The symbol $s \in \{1, \ldots, S\}$ is the video index and $t \in \{1, \ldots, T_s\}$ is the frame index.

To capture temporal continuity, we define a temporal Region of Interest (ROI) extractor $R(\cdot,\cdot)$ to highlight the temporal relationships between consecutive frames. Ideally, the temporal RoI derived from predictions should be consistent with that in the ground truth space. Thus, the temporal consistency at frame t of the s-th video sequence is expressed as:

$$\begin{aligned}\mathrm{TC}_s(t) &= Prob(R(P_s(t-1), P_s(t)), R(G_s(t-1), G_s(t))) \\ &\triangleq \frac{R(P_s(t-1), P_s(t)) \cap R(G_s(t-1), G_s(t))}{R(P_s(t-1), P_s(t)) \cup R(G_s(t-1), G_s(t))}\end{aligned} \tag{18}$$

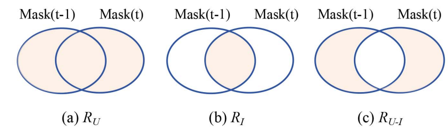


Fig. 9. Illustration of different temporal ROI extractor $R \in \{R_U, R_I, R_{U\text{-}I}\}$. (a) R_U represents the union between consecutive masks. (b) R_I denotes the intersection between consecutive masks. (c) $R_{U\text{-}I}$ indicates the difference between union and intersection of consecutive masks.

where $Prob(\cdot,\cdot)$ measures the consistency of temporal ROIs between the prediction and the ground truth spaces, implemented as mIoU in our work.

In order to integrate spatial accuracy and temporal continuity into a unified framework, we define the spatiotemporal consistency score:

$$\text{STC-score} = \frac{1}{S}\sum_{s=1}^{S}\frac{1}{T_s-1}\sum_{t=2}^{T_s}\mathrm{TC}_s(t)\cdot\mathrm{SC}_s(t) \tag{19}$$

where S is the total number of video sequences in the dataset and T_s represents the length of the s-th video sequence. This formulation incorporates both temporal and spatial consistency to provide a holistic assessment of video-based predictions.

Here, by substituting different $R \in \{R_U, R_I, R_{U\text{-}I}\}$ functions, as shown in Fig. 9, we define three specialized spatiotemporal consistency metrics: spatiotemporal coverage consistency ($\mathrm{STC}_{\mathrm{cover}}$) measures the temporal continuity of overall coverage using R_U, spatiotemporal alignment consistency ($\mathrm{STC}_{\mathrm{align}}$) evaluates the persistence of aligned regions over time with R_I, and spatiotemporal motion consistency ($\mathrm{STC}_{\mathrm{motion}}$) captures the coherence of motion with $R_{U\text{-}I}$, respectively. These three metrics are generally invariant to object size but sensitive to motion speed. In scenarios involving rapid motion with no overlap between consecutive frames, the $\mathrm{STC}_{\mathrm{align}}$ may fail due to its reliance on region alignment across frames, and $\mathrm{STC}_{\mathrm{cover}}$ becomes equivalent to $\mathrm{STC}_{\mathrm{motion}}$ since the captured coverage essentially reflects the object's movement.

C. Comparison With State-of-the-Arts

Baselines: To evaluate the effectiveness of the proposed MRCNet, we compare it with state-of-the-art methods from two categories: (i) Image camouflaged object detection: these approaches analyze static visual features in a single image to capture subtle differences between camouflaged objects and their environments. Prominent methods include SINet [1], SINet-v2 [23], ZoomNet [27], FSPNet [19], PUENet [56], HGINet [20] and CamoFormer [21]. (ii) Video object segmentation: These methods exploit spatiotemporal correlations to segment foreground objects in video sequences. The compared methods include MG [13], SLT-Net [15], IMEX [34], TSP-SAM [17], and ZoomNeXt [28].

Quantitative comparisons: We evaluate the proposed MRCNet alongside various baselines on the MoCA-Mask and CAD2016 benchmarks. From Table I, we can draw several key conclusions: (i) the substantial performance gap between the proposed MRCNet and image-based methods underscores

TABLE I
QUANTITATIVE COMPARISONS OVER TWO DATASETS. ↑/↓ INDICATES THAT LARGER/SMALLER IS BETTER. THE TOP THREE RESULTS ARE HIGHLIGHTED IN RED, GREEN, AND BLUE.

Method	Year	Input	MoCA-Mask						CAD2016					
			$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$\mathcal{M} \downarrow$	mDice↑	mIoU↑	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$\mathcal{M} \downarrow$	mDice↑	mIoU↑
SINet [1]	2020-CVPR	Image	0.574	0.185	0.655	0.030	0.221	0.156	0.636	0.346	0.775	0.041	0.381	0.283
SINet-v2 [23]	2021-TPAMI	Image	0.571	0.175	0.608	0.035	0.211	0.153	0.653	0.382	0.762	0.039	0.413	0.318
ZoomNet [27]	2022-CVPR	Image	0.582	0.211	0.536	0.033	0.224	0.167	0.633	0.349	0.601	0.033	0.349	0.273
FSPNet [19]	2023-CVPR	Image	0.594	0.182	0.608	0.044	0.238	0.167	0.681	0.401	0.716	0.044	0.446	0.332
PUENet [56]	2023-TIP	Image	0.594	0.204	0.619	0.037	0.302	0.212	0.691	0.485	0.795	0.034	0.514	0.396
HGINet [20]	2024-TIP	Image	0.506	0.105	0.615	0.042	0.137	0.086	0.568	0.289	0.718	0.063	0.344	0.233
CamoFormer [21]	2024-TPAMI	Image	0.684	0.404	0.790	0.017	0.437	0.362	0.717	0.524	0.831	0.036	0.551	0.458
MG [13]	2021-ICCV	Video	0.547	0.165	0.537	0.095	0.197	0.141	0.594	0.336	0.691	0.059	0.368	0.268
SLT-Net [15]	2022-CVPR	Video	0.656	0.357	0.785	0.021	0.387	0.310	0.696	0.481	0.845	0.030	0.493	0.401
IMEX [34]	2024-TMM	Video	0.661	0.371	0.778	0.020	0.409	0.319	0.695	0.486	0.845	0.030	0.498	0.407
TSP-SAM [17]	2024-CVPR	Video	0.689	0.444	0.808	0.008	0.458	0.388	0.751	0.628	0.851	0.021	0.603	0.496
ZoomNeXt [28]	2024-TPAMI	Video	0.734	0.476	0.736	0.010	0.497	0.422	0.757	0.593	0.865	0.020	0.599	0.510
SRRNet [57]	2025-ICCV	Video	0.727	0.489	0.803	0.008	0.513	0.428	0.723	0.541	0.553	0.025	0.553	0.452
EMIP [58]	2025-TIP	Video	0.675	0.381	-	0.015	0.426	0.333	0.719	0.514	-	0.028	0.536	0.425
MRCNet(Point)	Ours	Video	0.730	0.518	0.827	0.009	0.541	0.456	0.762	0.627	0.902	0.021	0.638	0.532
MRCNet(Box)	Ours	Video	0.741	0.530	0.826	0.008	0.556	0.472	0.777	0.651	0.899	0.019	0.663	0.562

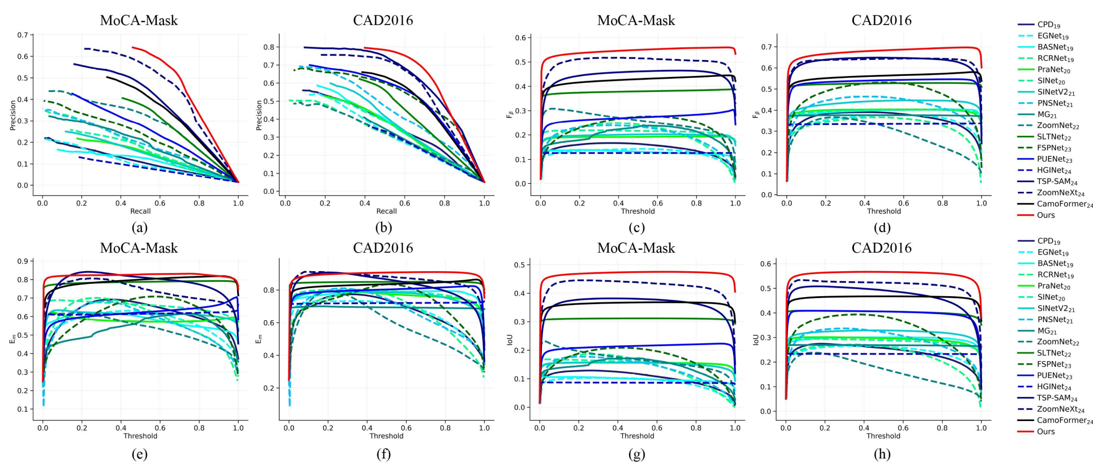


Fig. 10. The PR, F_β, E_m, and IoU curves of the proposed model and the recent SOTA algorithms over two datasets. (a) and (b) are precision-recall curves. (c) and (d) represent F_β curves. (e) and (f) are E_m curves. (g) and (h) indicate IoU curves.

the efficacy of motion representation learning based on the motion reasoning chain. (ii) Compared with video-based methods, MRCNet surpasses all baselines in six general metrics, achieving state-of-the-art performance. Notably, it outperforms ZoomNext [28] by 5.9% in terms of the mDice metric on the MoCA-Mask dataset. Meanwhile, compared to the TSP-SAM method based on prompt learning, MRCNet achieves a gain of 9.8% in the mDice metric. This improvement can be attributed to the promotion of the prior semantic knowledge extracted from the motion reasoning chain to the motion perception and visual comprehension of camouflaged objects, thereby improving the accuracy of prompts. (iii) The box prompts outperform point prompts by providing more complete object coverage, leading to higher detection precision.

Besides, PR, F_β, E_m, and IoU curves marked by red color in Fig. 10 further validate the effectiveness of the proposed MRCNet. The flatness of F_β, E_m, and IoU curves reflects more consistent and unified predictions. Moreover, the number of trainable parameters is shown in Table II. Compared with the state-of-the-art ZoomNeXt [28], MRCNet achieves substantial gains with a slight increase in trainable parameters.

TABLE II
COMPARISONS ON THE NUMBER OF PARAMETERS

Method	Trainable params(M)
FSPNet	274.24
HGINet	395.11
SLTNet	82.41
TSP-SAM	89.78
ZoomNext	84.78
MRCNet(Ours)	87.93

Qualitative comparisons: To provide more intuitive evaluations, Fig. 11 visualizes the prompts of our MRCNet and the baseline TSP-SAM. It is evident that compared with TSP-SAM,

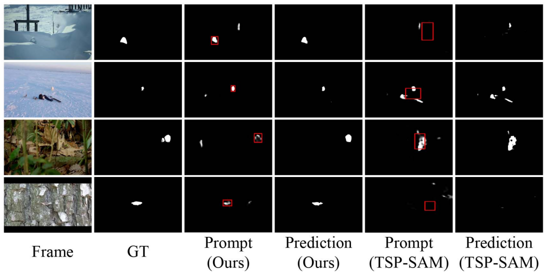


Fig. 11. Visualization of our proposed MRCNet and TSP-SAM with box prompts. From left to right: frames (column 1), ground truth (column 2), prompts of our MRCNet (column 3), prediction of our MRCNet (column 4), prompts of TSP-SAM (column 5), prediction of TSP-SAM (column 6).

the prompts of our MRCNet are less disturbed by the background, verifying the role of semantic knowledge in promoting the accuracy of prompts. In addition, we present segmentation results of several typical examples in Fig. 12. Our MRCNet effectively handles various complex scenes, including small objects (Row 1-2), middle objects (Row 3-7), big objects (Row 8), multiple objects (Row 8), low light (Row 3, 5, 8), and background interference (Row 1, 3).

D. Spatiotemporal Consistency Analysis

To qualitatively assess the tracking stability, we visualize the center trajectory of the predicted mask sequences in Fig. 13. Compared with baselines, the trajectory of the proposed MRCNet aligns more closely with the ground truth, exhibiting smaller deviations. Taking the arctic fox video sequence in Fig. 13(a) as an example, the baseline methods, including SLTNet [15] (frames 11, 13, and 23), TSP-SAM [17] (frames 11, 13, and 24), and ZoomNeXt [28] (frames 12, 23, and 24), display varying degrees of failure in tracking the camouflaged object. In contrast, our proposed MRCNet consistently delivers stable and accurate tracking across all frames.

For a quantitative assessment of tracking accuracy, we plot mIoU curves over frames in Fig. 14. The observations are as follows: (i) Image-based methods including PUENet and CamoFormer struggle with tracking over multiple frames due to the absence of motion modeling. In comparison, the proposed MRCNet maintains consistently higher mIoU, proving the crucial role of motion reasoning in breaking camouflage. (ii) compared to video-based baselines, such as SLTNet, TSP-SAM, and ZoomNeXt, the proposed MRCNet maintains its mIoU above the threshold across entire video sequences, reflecting successful tracking. For example, in the arctic fox sequence, MRCNet exhibits minimal fluctuations, whereas other methods show sudden drops, highlighting their struggle in maintaining temporal consistency. These analyses demonstrate that the proposed MRCNet accurately models the motion between frames driven by the motion reasoning chain, maintaining spatiotemporal consistency in video sequences.

To further evaluate the spatiotemporal continuity, we leverage the proposed spatiotemporal consistency metric to compare MRCNet with baselines. From Fig. 15, we can observe that MRCNet outperforms all baselines in terms of region coverage, region alignment, and motion coherence. Meanwhile, we plot $\mathrm{STC}_{\mathrm{cover}}$ curves of several video sequences to analyze the spatiotemporal continuity of overall coverage. As presented in Fig. 16, the curve of our proposed MRCNet consistently remains above those of the baseline methods, demonstrating superior temporal consistency across various scenarios. For instance, in sequences like (a) arctic fox and (c) flower crab spider 0, MRCNet maintains a high and stable $\mathrm{STC}_{\mathrm{cover}}$ score, indicating smooth and coherent motion predictions. In contrast, baselines such as PNS-Net and SLTNet exhibit significant fluctuations and lower scores, suggesting less stable spatiotemporal coverage. Moreover, in challenging cases like (g) snow leopard 10, where baseline methods struggle with highly dynamic backgrounds and abrupt motions, MRCNet still achieves a relatively better performance, highlighting its robustness in handling complex motion patterns.

E. Ablation Study

Ablation study of the proposed modules: To verify the effectiveness of each component, we conduct ablation studies by progressively adding motion representation learning (MRL), cross modal prompt learning (CPL), and semantic knowledge injection (SKI). As shown in Table III, each module yields consistent gains, demonstrating that the motion reasoning chain effectively enhances motion perception and visual comprehension of camouflaged objects.

Ablation study of MLLMs for box prompting SAM: We conduct comparative experiments by using MLLMs to predict prompts for SAM. As reported in Table IV, this strategy performs noticeably worse than MRCNet, as the inherent difficulty in camouflaged scenes makes precise prompt generation difficult for MLLMs. We further visualize the prompts predicted by MLLMs and our MRCNet. From Fig. 17, we can find that the prompts predicted by Qwen-VL [40] and Pink [46] exhibit noticeable deviations in size or position, while our MRCNet yields better-aligned prompts.

Ablation study of VFMs: To assess the impact of VFMs, we compare SAM and SAM2 in Table IV. Obviously, SAM2 performs better than SAM on CAD2016 but worse on MoCA-Mask, which is attributed to the higher sensitivity of SAM2 to the reliability of prompts. CAD2016 features mainly medium-sized objects with weak camouflage, learning more consistent prompts. In contrast, MoCA-Mask contains small objects and challenging camouflage, leading to deviated prompts frequently. This makes SAM2 more susceptible to error accumulation during mask propagation, leading to inferior performance compared to SAM.

Ablation study of the motion reasoning chain: To verify the effectiveness of the motion reasoning chain, we conduct ablation studies on the selection of MLLMs and the reasoning steps. From Table V, comparable performance is observed for Qwen-VL [40] and InternVL [41] against Video-LLaVA [49], which is consistent with the homogeneity of MLLMs [59]. This indicates that performance depends more on how MLLMs are used for reasoning and semantic guidance than on the specific model choice. Furthermore, Table VI shows that the full chained

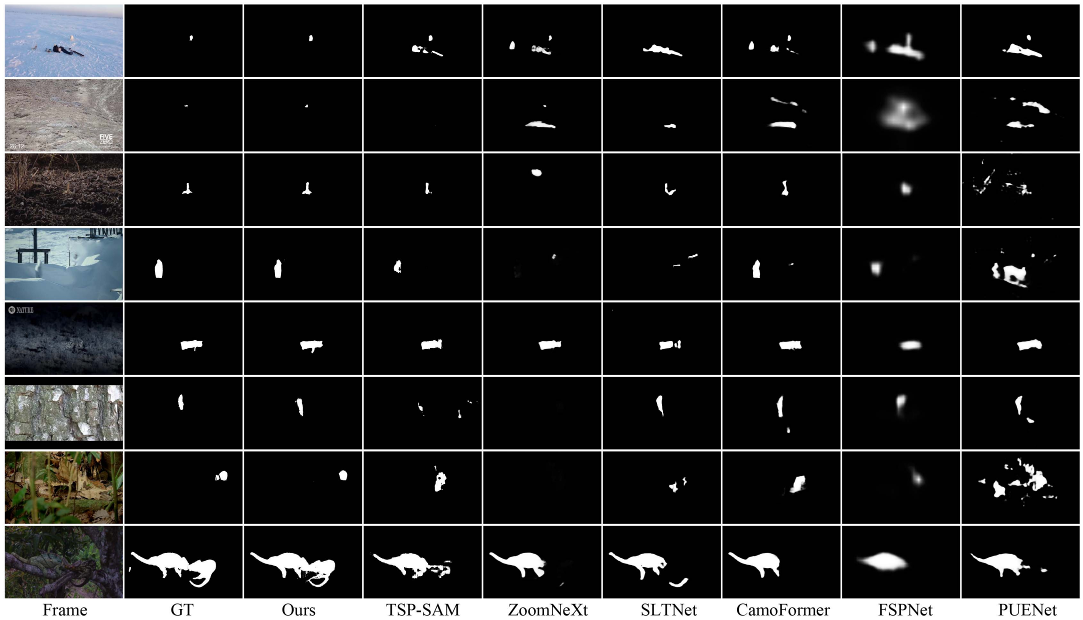


Fig. 12. Visualization of our proposed MRCNet and baseline methods over two datasets. From left to right: frames (column 1), ground truth (column 2), prediction of our MRCNet (column 3), prediction of compared methods (columns 4-9).

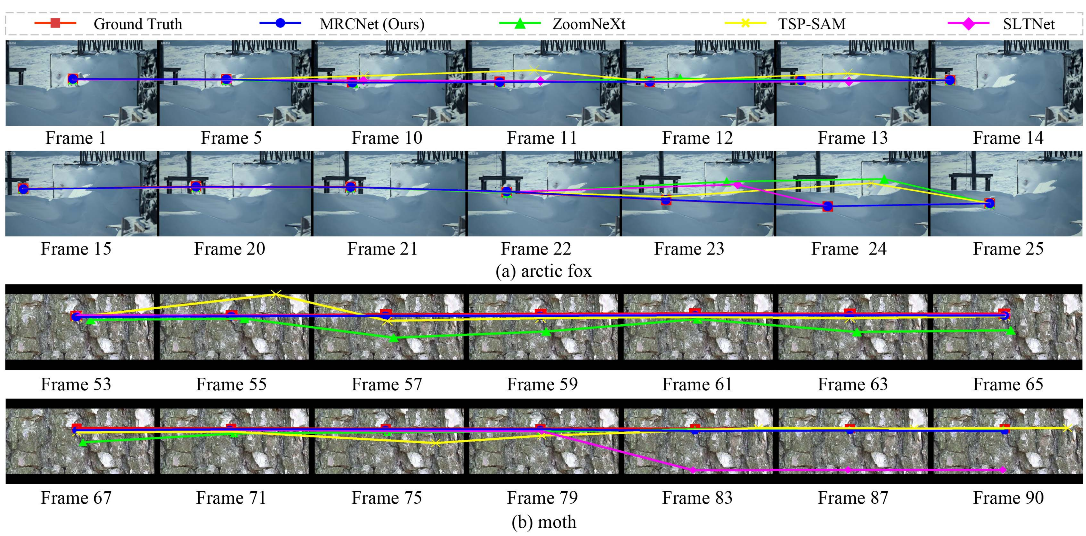


Fig. 13. Center trajectory of segmentation masks. (a) The arctic fox video sequence. (b) The moth video sequence.

TABLE III
ABLATION STUDIES ON THE PROPOSED MRCNET. MRL: MOTION REPRESENTATION LEARNING DRIVEN BY THE MOTION REASONING CHAIN. CPL: CROSS MODAL PROMPT LEARNING. SKI: SEMANTIC KNOWLEDGE INJECTION.

Settings	MRL	CPL	SKI	MoCA-Mask				CAD2016			
				$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mIoU↑	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mIoU↑
1	✗	✗	✗	0.700	0.473	0.791	0.425	0.725	0.566	0.861	0.487
2	✓	✗	✗	0.710	0.484	0.790	0.432	0.738	0.593	0.872	0.507
3	✗	✓	✗	0.719	0.511	0.819	0.457	0.754	0.604	0.882	0.526
4	✗	✗	✓	0.714	0.500	0.818	0.444	0.748	0.613	0.891	0.522
5	✓	✓	✗	0.725	0.521	**0.834**	0.466	0.773	0.619	0.887	0.538
6	✓	✗	✓	0.724	0.513	0.804	0.453	0.768	0.641	**0.904**	0.550
7	✗	✓	✓	0.729	0.528	0.829	0.466	0.775	0.623	0.880	0.537
8	✓	✓	✓	**0.741**	**0.530**	0.826	**0.472**	**0.777**	**0.651**	0.899	**0.562**

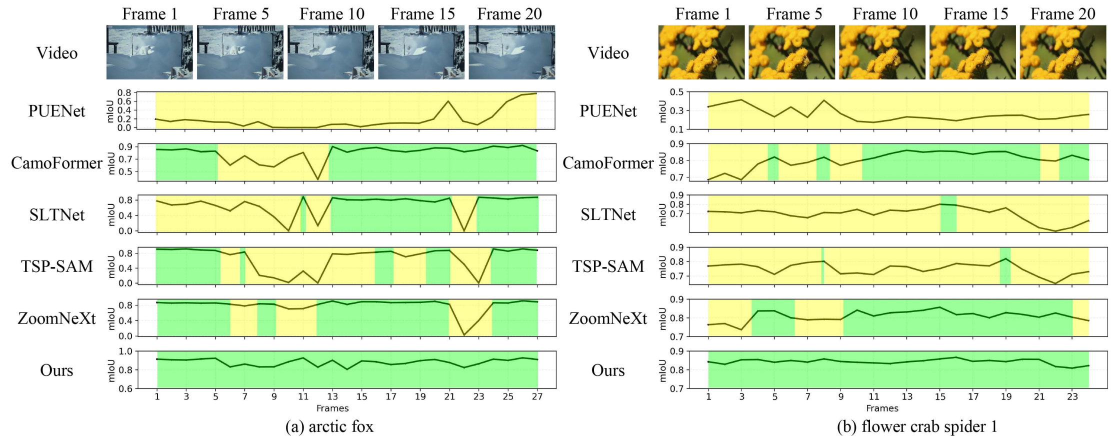


Fig. 14. Tracking performance of several typical video sequences. Empirically, 0.8 is taken as the threshold for mIoU. The green part denotes successful tracking, and the yellow part indicates failed tracking.

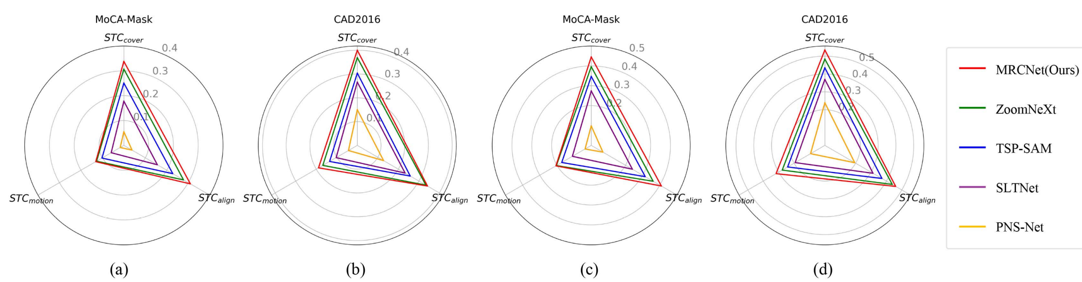


Fig. 15. Performance comparisons in terms of the proposed spatiotemporal consistency metric. (a) and (b) represent the performance comparisons at $Prob(\cdot,\cdot) = \mathrm{mIoU}(\cdot,\cdot)$ on the MoCA-Mask and CAD2016 datasets, respectively. (c) and (d) denote the performance comparisons at $Prob(\cdot,\cdot) = \mathrm{mDice}(\cdot,\cdot)$ on both datasets.

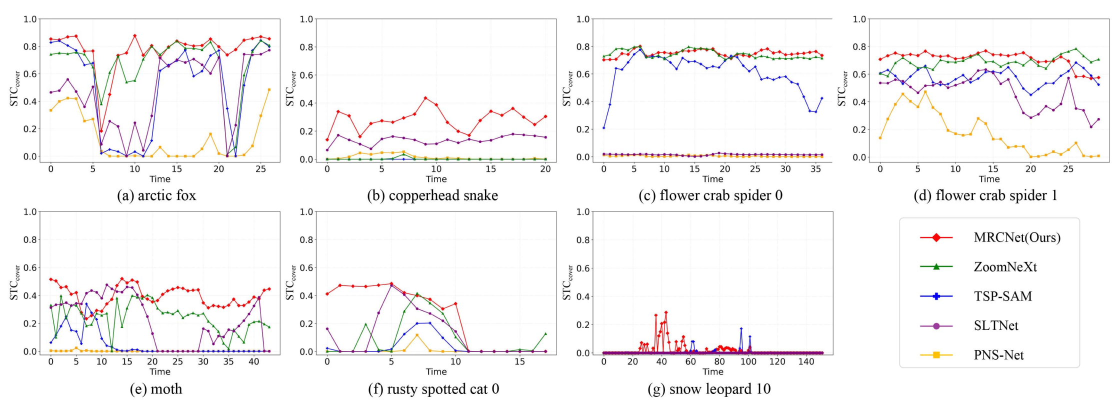


Fig. 16. The spatiotemporal coverage consistency ($\mathrm{STC_{cover}}$) curves of the proposed MRCNet and the compared baselines. The higher the $\mathrm{STC_{cover}}$ score, the better the spatiotemporal consistency of predictions.

TABLE IV
ABLATION STUDIES OF MLLMS FOR BOX PROMPTING VISUAL FOUNDATION MODELS ON TWO BENCHMARK DATASETS

MLLMs	VFMs	MoCA-Mask					CAD2016				
		$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mDice↑	mIoU↑	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mDice↑	mIoU↑
Qwen-VL [40]	SAM [16]	0.698	0.479	0.821	0.490	0.407	0.705	0.551	0.815	0.547	0.448
Pink [46]	SAM [16]	0.510	0.265	0.553	0.288	0.242	0.535	0.442	0.611	0.456	0.386
MRCNet (Ours)	SAM [16]	**0.741**	**0.530**	0.826	**0.556**	**0.472**	0.777	0.651	0.899	0.663	0.562
Qwen-VL [40]	SAM2 [39]	0.688	0.444	0.815	0.447	0.381	0.788	0.697	0.895	0.687	0.582
Pink [46]	SAM2 [39]	0.490	0.264	0.539	0.297	0.244	0.548	0.480	0.616	0.485	0.426
MRCNet (Ours)	SAM2 [39]	0.709	0.478	**0.827**	0.492	0.433	**0.826**	**0.746**	**0.945**	**0.746**	**0.654**

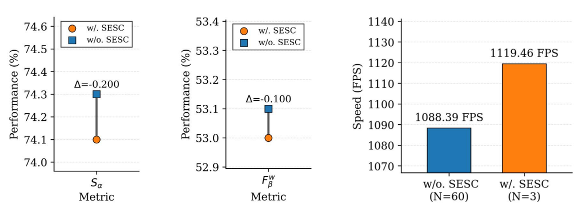
TABLE V
ABLATION STUDIES OF THE MLLMS IN THE MOTION REASONING CHAIN. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

MLLMs	MoCA-Mask					CAD2016				
	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mDice↑	mIoU↑	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mDice↑	mIoU↑
Qwen-VL [40]	0.739	0.525	**0.832**	0.548	0.465	0.773	0.641	0.891	0.653	0.552
InternVL [41]	**0.742**	**0.530**	0.825	0.553	0.469	0.775	0.641	0.894	0.655	0.554
Video-LLaVA [49]	0.741	**0.530**	0.826	**0.556**	**0.472**	**0.777**	**0.651**	**0.899**	**0.663**	**0.562**

TABLE VI
ABLATION STUDIES OF MULTIPLE REASONING STEPS IN THE MOTION REASONING CHAIN

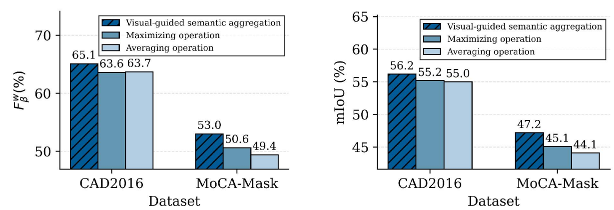
Reasoning strategy	Settings	MoCA-Mask					CAD2016				
		$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mDice↑	mIoU↑	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	mDice↑	mIoU↑
-	w/o. MRC	0.700	0.473	0.791	0.503	0.425	0.725	0.566	0.861	0.594	0.487
Individual	Step 2	0.728	0.520	0.819	0.543	0.459	0.774	0.643	0.899	0.658	0.555
	Step 2 + Step 3	0.731	0.528	0.821	0.549	0.465	0.774	0.645	0.900	0.659	0.555
Chained	Step 1 + Step 2	0.732	0.526	**0.826**	0.552	0.466	0.775	0.651	**0.904**	0.661	0.558
	Step 1 + Step 2 + Step 3	**0.741**	**0.530**	**0.826**	**0.556**	**0.472**	**0.777**	**0.651**	0.899	**0.663**	**0.562**

Fig. 17. Visualization of the prompts produced by MLLMs and our MRCNet.



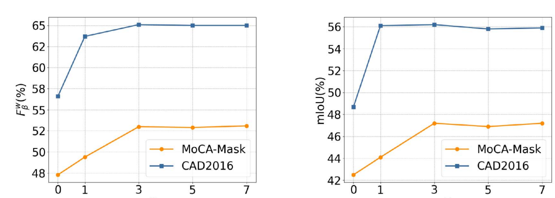
(a) S_α metric (b) F_β^w metric (c) FPS metric

Fig. 18. Ablation results of the self-evolving semantic condensation (SESC). (a)–(b) Performance comparisons. (c) Comparisons of the inference speed.



(a) F_β^w metric (b) mIoU metric

Fig. 19. Ablation on the visual-guided semantic aggregation.



(a) F_β^w metric (b) mIoU metric

Fig. 20. Ablation study on the number of representative semantic descriptors K over two datasets.

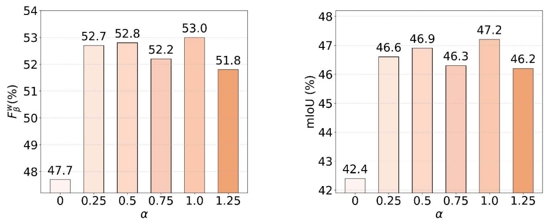
reasoning achieves the best results, underscoring the importance of the chained design and its reasoning order for stable and effective results.

Ablation study of de-biased motion prototype learning: To examine the effect of de-biased motion prototype learning, we conduct ablation studies from two perspectives. From Fig. 18, it can be observed clearly that the accuracy is largely preserved while inference speed is improved when self-evolving semantic condensation is enabled, indicating its effectiveness in filtering semantic redundancy. Moreover, as shown in Fig. 19, replacing visual-guided semantic aggregation with averaging or maximizing causes a clear performance drop, confirming its importance in suppressing MLLM hallucinations.

Sensitivity analysis of hyper-parameters: We analyze the sensitivity of MRCNet to two key hyper-parameters. As presented in Fig. 20, the segmentation performance peaks at $K = 3$, indicating that three representative semantic descriptors are sufficient to mitigate MLLM hallucinations. In addition, Fig. 21 presents the sensitivity analysis on the loss balance factor α. The best performance is achieved at $\alpha = 1$.

F. Camouflaged People Detection

To assess the generalization capability, we test MRCNet on the camouflaged people detection dataset [54]. As reported in Table VII, the proposed MRCNet outperforms all baselines, achieving state-of-the-art performance. This demonstrates that the proposed MRCNet is able to generalize unseen data more effectively. Additionally, Fig. 22 further provides visualizations for qualitative evaluation. Compared with baselines, MRCNet



(a) F_β^w metric (b) mIoU metric

Fig. 21. Ablation study on the balance factor α in the loss function over MoCA-Mask dataset.

TABLE VII
GENERALIZATION COMPARISONS ON THE CPD DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**.

Method	$S_\alpha \uparrow$	$F_\beta^w \uparrow$	$E_\phi \uparrow$	$\mathcal{M} \downarrow$	mDice↑	mIoU↑
SINet [1]	0.656	0.259	0.672	0.025	0.319	0.243
SINet-v2 [23]	0.724	0.453	0.805	0.013	0.488	0.393
FSPNet [19]	0.618	0.200	0.620	0.037	0.255	0.182
PUENet [56]	0.673	0.372	0.724	0.022	0.422	0.324
HGINet [20]	0.502	0.108	0.548	0.052	0.144	0.092
CamoFormer [21]	0.789	0.601	0.874	0.007	0.616	0.522
SLT-Net [15]	0.539	0.151	0.732	0.023	0.165	0.113
TSP-SAM [17]	0.723	0.511	0.819	0.011	0.513	0.432
ZoomNeXt [28]	0.747	0.544	0.775	0.008	0.539	0.448
MRCNet (Point)	0.803	0.654	**0.895**	**0.007**	0.661	0.579
MRCNet (Box)	**0.806**	**0.656**	0.888	**0.007**	**0.664**	**0.587**

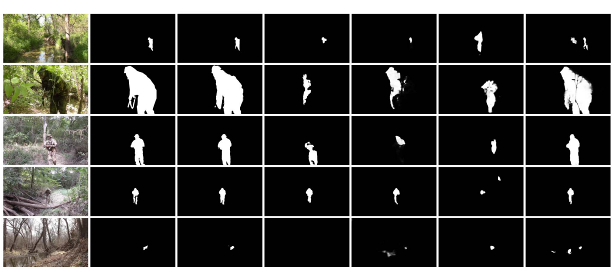


Fig. 22. Visualization of the proposed MRCNet and baselines on the CPD dataset.

enables accurate detection of camouflaged objects and superior robustness against varying backgrounds.

V. Conclusion and Future Work

This paper presents MRCNet, a cross-modal VCOD framework that imitates the human thought process when observing camouflaged objects. By introducing an MLLM-based generative sampling strategy, the motion reasoning chain is developed to reason about motion and concept attributes of camouflaged objects for visual comprehension. To facilitate the identification of camouflaged objects, motion representation learning is proposed that uses the de-biased motion prototype learning strategy to suppress hallucination-induced semantic drifts, thereby enhancing motion perception. To learn precise prompts for the visual foundation model, cross modal prompt learning integrates the de-biased concept prototype into visual representations to promote the visual comprehension of camouflaged objects. Extensive experiments verify the effectiveness of the proposed method on the general metrics and our proposed spatiotemporal consistency metrics.

In future work, we will further expand the motion reasoning chain strategy with more diverse motion knowledge, including subtle dynamic patterns, interactions between moving objects, etc. Additionally, we will investigate the applicability of this motion reasoning strategy to broader video tasks.

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